Computer Organization & Assembly Lang.

Buses

★ Bus:-

A bus, in computing and digital technology, is electronic pathway through which data can be transferred.

Types:-

(1) Data Bus (2) Address Bus (3) Control Bus

→ Data Bus and Control Bus are bidirectional while on the other hand Address Bus is unidirectional.

Registers

★ In CPU (Central Processing Unit) registers are used to perform any operation.

* Buses are used to fetch the data from RAM to CPU

* Index / Pointer / Base Register.
    - update memory address

* PC Register.-
    - PC stands for program counter.
    - These registers give address instruction that where to break the sequence.
    - it breaks the sequence if the condition is not fulfiled.

★ Flag Register :-

Flag register include flags that tell us about program. Flag register is of 8 bits.

Every bit of flag register is dealt separately and every bit have a different purpose.

★ Mnemonic :-

A mnemonic in computer programming is a shortcut or a trick that helps you to remember complex instructions or commands.

We can say that it is a mixture of high-level language and assembly language.

★ Instructional Loops:-

(1) Data instruction (2) Arithmatic & Logical instructions
(3) Program Control instructions (4) Special instruction

$\boxed{\text{Note}}$ Special instructions can change the behavior of CPU.

★ General Purpose Register:-

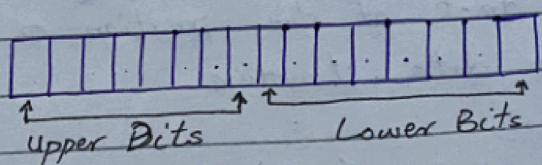
General purpose registers are used calculate data (addition, subtraction etc) and to save addresses.

i) AX → Accumulator Register → 16 Bit
ii) BX → Base Register → 16 Bit
iii) CX → Counter Register → 16 Bit
iv) DX → Data Register → 16 Bit

- They can perform different tasks rather than their name.
- AX is used for arithmatic operations.
- BX is used for starting a code.
- The name of general purpose registers can be changed according to processor.
- These are 16 Bit registers but we can use them as 8 Bit and also 32 Bit.
- If we want to use upper bits, we will use AH and to use lower bits we will use AL.

Example:-

16 Bit



• If we extend the 16 bit to 32bit the names of the registers will be changed.

New Name

EAX → Extended Accumulator Register → 32 Bit

EBX → Extended Base Register → 32 Bit

ECX → Extended Counter Register → 32 Bit

EDX → Extended Data Register → 32 Bit

★ Index :-

An index refers to a register that is used to keep track of the position or location of data within a data structure.

Types:-

(1) Source Index (SI) (2) Destination Index (DI)

PC Register → Program Counter Register
IP Register → Instruction Pointer Register
SP → Stack Pointer
BP → Base Pointer
* Segment Registers:-
CS → Code Segment Register
DS → Data Segment Register
SS → Stack Segment Register
EX → Extra Segment Register